

Takeaway: Coupled Axial Green Solvers

GreenNet learns line-wise Green kernels; CouplingNet learns the directional source split.

A 2D elliptic solve is represented as axial Green inversions plus a learned, balance-preserving source decomposition.

AXIAL GREEN KERNELS

Learn unit-interval line-wise inverse kernels.

ANALYTIC STRUCTURE

Build singular Green behavior before learning smooth correction.

SOURCE SPLIT

Learn a balance-preserving ϕ/ψ decomposition.

ENERGY BOUND

Control final solution error through split-energy consistency.

Axial Green operators provide the line-wise inverses, and CouplingNet supplies the missing source coupling.